Mudassir Shah

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Date of birth: 1996/03/02 **Place of birth:** Pakistan **Nationality:** Pakistani

ABOUT ME

Computational spatial omics researcher developing deep learning methods for spatial metabolomics and spatial proteomics. Experienced in multimodal spatial data integration, MSI–histology fusion, learning based tissue phenotyping, with focus on modeling tumor microenvironment organization and molecular heterogeneity.

WORK EXPERIENCE

Shanghai Jiao Tong University

City: Shanghai | **Country:** China

[2025/08/01 – Current] **Postdoctoral Researcher**

Developing deep learning methods for multimodal spatial omics integration, including MSI–histology registration, spatial proteomics alignment, and AI-guided tissue phenotyping for tumor microenvironment analysis.

National Institute for Data Science in Health and Medicine, Xiamen University

City: Xiamen | **Country:** China

[2020/09/06 – 2025/06/26] **Doctoral Researcher (PhD Candidate)**

Conducted independent research in computational spatial metabolomics and proteomics.

Developed deep learning frameworks (SagMSI, eLIMS, MSInet, MSIreg).

Published multiple first-author papers in top journals (Analytica Chimica Acta, J. Proteome Research, Analytical Chemistry).

Supervised MSc students and contributed to teaching in computational biology.

The Aga Khan Health Service

City: Karachi | **Country:** Pakistan

[2018/03/28 – 2018/08/10] **Data Analyst – Research & Healthcare**

Analyzed electronic health records and patient data.

Designed databases to manage and retrieve large-scale patients and hospital records.

Software Demonstration and Training

Department of Computer Science, Hazara University Mansehra

City: Mansehra | **Country:** Pakistan

[2017/09/17 – 2018/03/15] **Teaching Intern (Lecturer role)**

Computer Teacher on Matlab and R.

Logic Gates Designing and Control System.

EDUCATION AND TRAINING

[2020/09/06 – 2025/06/26] **PhD in Electronic Science and Technology**

Xiamen University https://www.xmu.edu.cn/

City: Xiamen | **Country:** China

[2018/09/11 – 2020/07/14] **M.S. in Electronic Science and Technology**

University of Electronic Science and Technology of China https://en.uestc.edu.cn/

City: Chengdu | **Country:** China

[2013/09/15 – 2017/07/22] **B.S.Hons. in Electronic Engineering**

Islamia College University Peshawar https://www.icp.edu.pk/

City: Peshawar | **Country:** Pakistan

SKILLS

Computer Vision | Machine Learning | Biomedical and Molecular Imaging | Bioinformatics | Image Processing | Multi-Modal Data Integration | Artificial Intelligence and Neural Networks | Mass Spectrometry Imaging data and its interpretation

PUBLICATIONS

[2025] **MSInet: A Self-Supervised CNN Framework Integrating Global and Local Context for Robust Mass Spectrometry Imaging Segmentation**

Self-supervised deep learning framework for spatial tissue phenotyping in mass spectrometry imaging, capturing tumor microenvironment structure through spatial feature learning.

Authors: Mudassir Shah, Siyang Liu, Lei Guo, Zhang Yusong, Xiangnan Xu, Fei, Zhaodong, Jingjing Xu, Jiyang Dong | **Journal Name**: Analytical Chemistry | **Volume, Issue and Pages**: 97, 44, 24697–24705

[2025] **SagMSI: A Graph Convolutional Network framework for Precise Spatial Segmentation in Mass Spectrometry Imaging**

Developed SagMSI, an unsupervised graph convolutional network (GCN) model for spatial segmentation of MSI data. The method integrates novel spatially aware graph construction with GCN-based feature learning, enabling precise identification of tissue substructures and tumor margins revealing biologically meaningful sub-regions and potential biomarker ions.

Authors: Mudassir Shah, Linlin Wang, Lei Guo, Chengyi Xie, Thomas Ka-Yam Lam, Lingli Deng, Xiangnan Xu, Jingjing Xu, Jiyang Dong, Zongwei Cai | **Journal Name**: Analytica Chimica Acta | **Volume, Issue and Pages**: Volume 1358, 2025, 344098, ISSN 0003-2670

[2024] **eLIMS: Ensemble Learning-Based Spatial Segmentation of Mass Spectrometry Imaging to Explore Metabolic Heterogeneity**

Developed an ensemble learning framework for spatial segmentation of mass spectrometry imaging data. The method integrates randomized UMAP-based dimensionality reduction with ensemble clustering to robustly capture subtle metabolic patterns, offering new insights into metabolic heterogeneity.

Authors: Mudassir Shah, Lei Guo, Xiangnan Xu, Lingli Deng, Keyi Lu, Jiyang Dong, Chao Zhao,* and Jingjing Xu | **Journal Name**: Journal of Proteome Research | **Volume, Issue and Pages**: 23 (8), 3088-3095

[2023] **Multimodal Image Fusion Offers Better Spatial Resolution for Mass Spectrometry Imaging**

Co-authored DeepFERE, a deep learning framework that reregister and fuse MSI with microscopy and spatial transcriptomics for high-resolution reconstruction. Improved tumor boundary detection and preserved chemical information.

Authors: Lei Guo, Jinyu Zhu, Keqi Wang, Kian-Kai Cheng, Jingjing Xu, Liheng Dong, Xiangnan Xu, Can Chen, Mudassir Shah, Zhangxiao Peng, Jianing Wang, Zongwei Cai, and Jiyang Dong | **Journal Name**: Analytica Chemistry | **Volume, Issue and Pages**: 95 (25), 9714-9721

[2022] **iSegMSI: An Interactive Strategy to Improve Spatial Segmentation of Mass Spectrometry Imaging Data**

Co-developed iSegMSI, an interactive segmentation strategy that integrates user scribbles as constraints to refine unsupervised MSI segmentation, improving segmentation accuracy and tolerance to annotation noise.

Authors: Lei Guo, Xingxing Liu, Chao Zhao, Zhenxing Hu, Xiangnan Xu, Kian-Kai Cheng, Peng Zhou, Yu Xiao, Mudassir Shah, Jingjing Xu, Jiyang Dong, and Zongwei Cai | **Journal Name**: Analytica Chemistry | **Volume, Issue and Pages**: 94(42), 14522-14529

[2025] **prepIMS: a Robust Data Preprocessing Workflow for Ion Mobility Mass Spectrometry Imaging**

Co-authored prepIMS, a preprocessing framework for ion mobility-MSI using density cluster-based peak detection. Outperformed conventional methods under noise, enabling accurate detection of isomeric/isobaric ions and supporting reliable 4D spatial metabolomics analysis.

Authors: Linlin Wang, Chengyi Xie, Thomas Ka Yam Lam, Chris Kong Chu Wong, Mudassir Shah, Xiangnan Xu, Jianing Wang, Jingjing Xu, Jiyang Dong, Zongwei Cai | **Journal Name**: Analytica Chimica Acta | **Volume, Issue and Pages**: 1384,344951, 0003-2670

[2026] **IregMSI: Deep Learning Assisted Automatic Multimodal Registration of Mass Spectrometry Imaging**

Developing IregMSI, a deep learning–assisted framework for automatic multimodal registration of spatial metabolomics, proteomics and histological data, enabling accurate integration for downstream analysis in application to spatial biology.

Authors: Mudassir Shah, Yong Wei, Jonathan Li, Jingjing Xu, and Wang Liu | **Journal Name**: Analytical Chemistry | **Volume, Issue and Pages**: Under-review

[2026] **Self-Supervised Spatial Denoising for Mass Spectrometry Imaging Enhances Metabolite Identification and Image–Omics Integration**

Authors: Mudassir Shah, Yu Ling, Zaib Khan, Wei Chen, and Wang Liu | **Journal Name**: briefings in bioinformatics | **Volume, Issue and Pages**: in preparation

HONOURS AND AWARDS

[2017/07/28] **Gold Medalist B.S.Hons. Awarding institution:** Islamia College University, Peshawar Pakistan.

Gold Medal – Top of the Class (2013–2017)

Awarded the Gold Medal as the highest-ranking graduate of the 2013–2017 cohort in Electronic Engineering, recognizing outstanding academic performance, dissertation excellence, and overall distinction among peers.

[2020/08/05] **Chinese Government Scholarship (2020-2025) Awarding institution:** Xiamen University

Awarded under the Chinese Government Scholarship program to pursue doctoral studies at Xiamen University (2020–2025). Covered tuition, living stipend, and research expenses for doctoral studies to carry research in computational methods for mass spectrometry imaging and spatial metabolomics.

[2019/06/18] **UESTC Academic Award (2018-2019) Awarding institution:** University of Electronic Science and Technology of China

Academic excellence award at the University of Electronic Science and Technology of China. Granted for outstanding coursework performance and significant co-curricular contributions during the Master's degree.

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

Chinese

LISTENING B1 **READING** A1 **WRITING** A1

SPOKEN PRODUCTION B1 **SPOKEN INTERACTION** B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user